

Connecticut Housing Sales Analysis (2022)

Final Project – Written Description

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Summary:

For my final project I analyzed real residential home sales in Connecticut from 2022. I basically wanted to understand the housing market better so I looked at things like which towns are the most expensive, which ones have the most sales, whether the type of home changes the price, and what the overall price distribution looks like. Once I had the data loaded I cleaned it up and made some charts to explore it, then I got it ready for machine learning by converting the categorical columns into numbers and splitting everything into training and testing sets. After that I trained three different models like Linear Regression, Polynomial Regression, and a Decision Tree and compared them to see which one did the best job predicting sale prices based on assessed value, town, and residential type.

Methods:

Data Collection

I got my data from the U.S. government's open data catalog at catalog.data.gov. The specific dataset is called "Real Estate Sales 2001–2018" and contains official property sale records from the state of Connecticut. The whole dataset is really big and goes back to 2001 so I just used the filter option directly on the catalog.data.gov website to filter by List Year 2022 only and then I downloaded only that part as a CSV. That gave me 43,470 rows to work with, where each row represents one property sale in Connecticut in 2022.

Data Exploration

The first thing I did after loading the CSV was just look at it to figure out what I was working with. I ran `df.info()`, `df.head()`, and `df.describe()` to see the column types, check for missing values, and get a basic idea of the numbers. I also checked what kinds of properties were in the data and how many unique towns there were, it turned out there were 138 towns total. About 10% of the rows were missing a Residential Type which I knew I was going to have to deal with later.

Data Cleaning

The data was pretty messy so there was a lot of cleaning to do before I could use it. First, I dropped four columns that were either mostly empty or not going to help with predictions: "Non Use Code," "Assessor Remarks," "OPM remarks," and "Location." The Sale Amount and Assessed Value columns were stored as text strings with dollar signs and commas in them, like "\$250,000," so I had to strip those characters out using `str.replace()` and then convert both columns to actual numbers using `pd.to_numeric()`. After that, I filtered the data to only keep

residential properties because commercial and industrial sales are totally different and would mess up the model. I also cut out any sales under \$10,000 or over \$2,000,000 because those are probably not normal market sales. For the rows missing a Residential Type I filled them in with "Unknown" instead of just deleting them. Finally, I took a random sample of 4,000 rows using `random_state=42` so that the models would run in a reasonable amount of time and the results would be the same every time I ran the code.

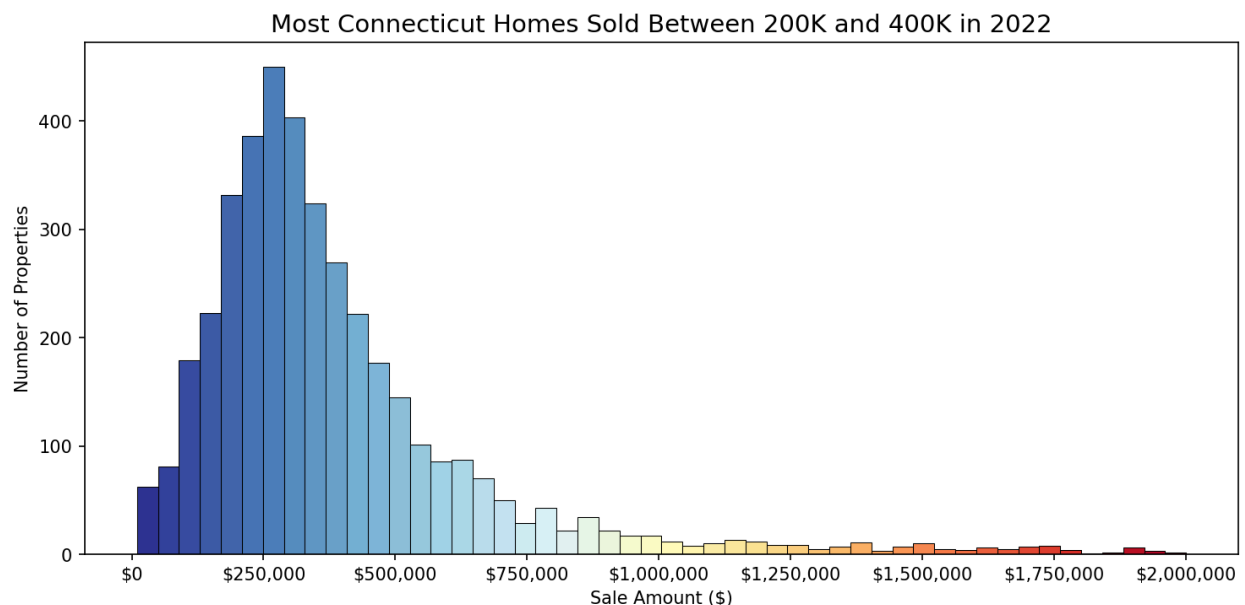
Exploratory Analysis

Before I got into the machine learning part, I made six charts to actually understand what the data looked like. The first one was a histogram of sale prices and it showed that most Connecticut homes sold somewhere between \$200K and \$400K which made sense. The second was a box plot that compared the price ranges for each type of home. I used this because it shows more than just the average it shows how spread out the prices are too. The third chart was a horizontal bar chart of the top 10 most expensive towns, towns like Westport and New Canaan were at the top. The fourth was a bar chart showing which towns had the most sales, so basically where people are actually buying homes the most. The fifth was a line chart that showed how average sale prices changed month by month throughout 2022, I wanted to see if there was a pattern or a best time to buy. The sixth and last chart was a pie chart breaking down what percentage of sales came from each residential type, which showed that single family homes make up by far the biggest share.

Results:

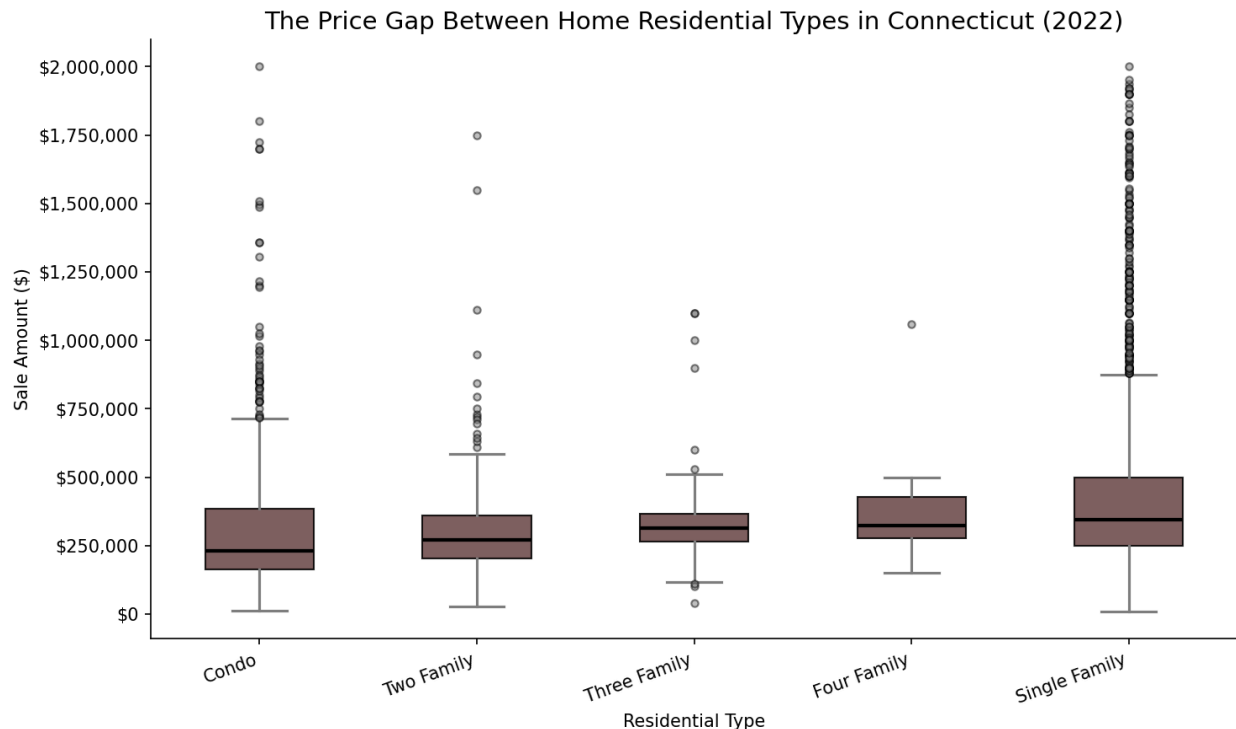
My results came from two parts, the charts I made and the models I trained.

Figure 1 - Price Distribution



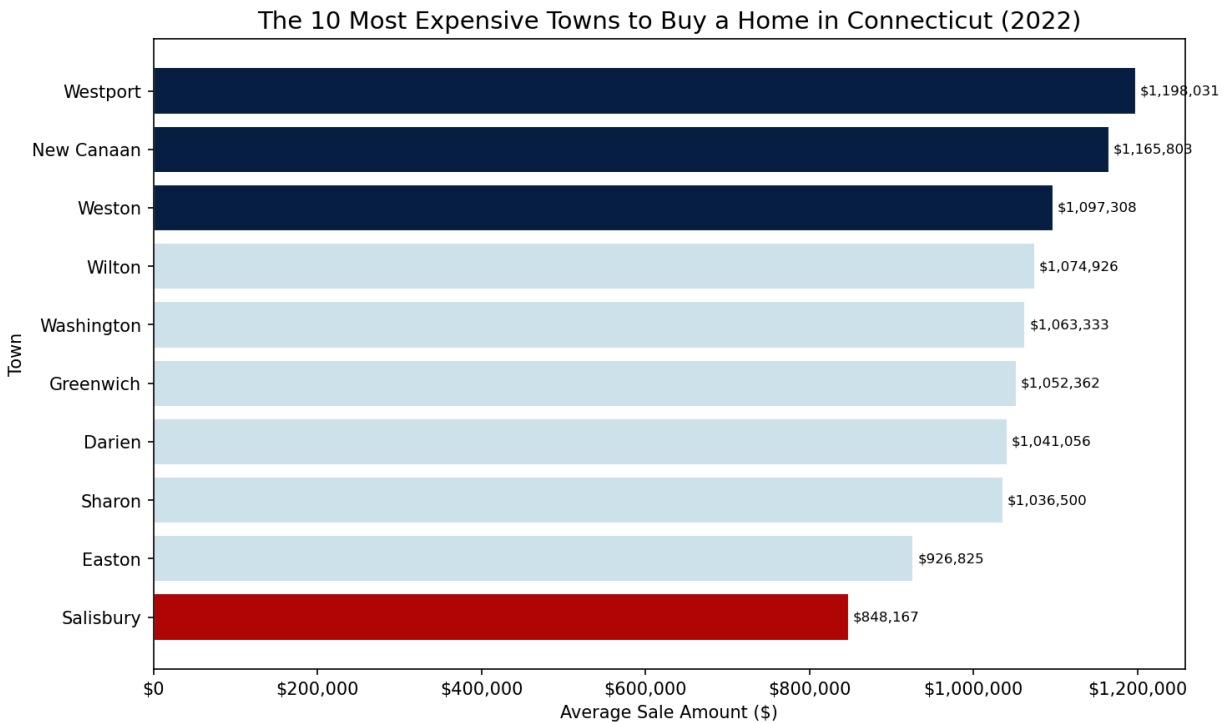
This histogram shows that most homes in Connecticut sold somewhere between \$200K and \$400K in 2022, which honestly was not that surprising. Most of the bars are bunched up on the left side and then it kind of just trails off, so there are way more cheaper homes being sold than expensive ones.

Figure 2 - Price by Residential Type



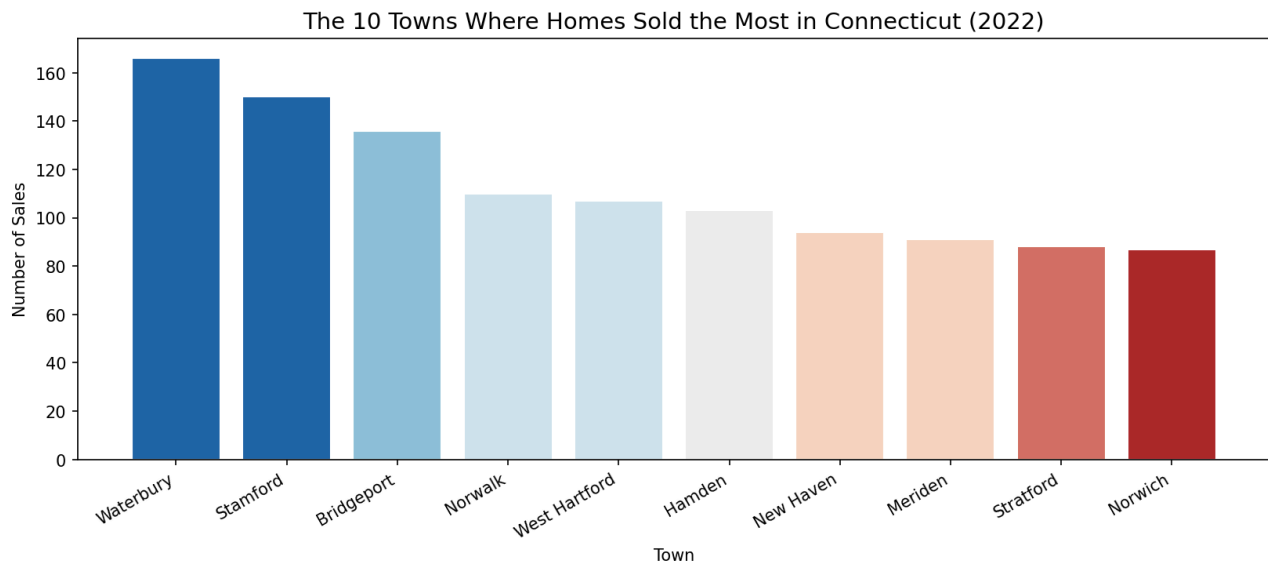
This is a box plot which shows the spread of prices for each home type. The box shows where most of the sales fell and the line in the middle of the box is the median price. The lines going up and down from the box show how far prices stretched above and below that middle range. The little dots floating above the lines are basically the really expensive outliers, like condos that sold for \$1.5M or \$2M which is way above what most condos go for. Looking at the chart, Condos had the lowest median price out of all the types which I did not expect at all. Single Family homes had the most outlier dots at the top which makes sense since there are just so many single family homes in Connecticut. But honestly all the medians were pretty close to each other so the type of home did not make as big of a difference on price as I thought it would.

Figure 3 - Top 10 Most Expensive Towns



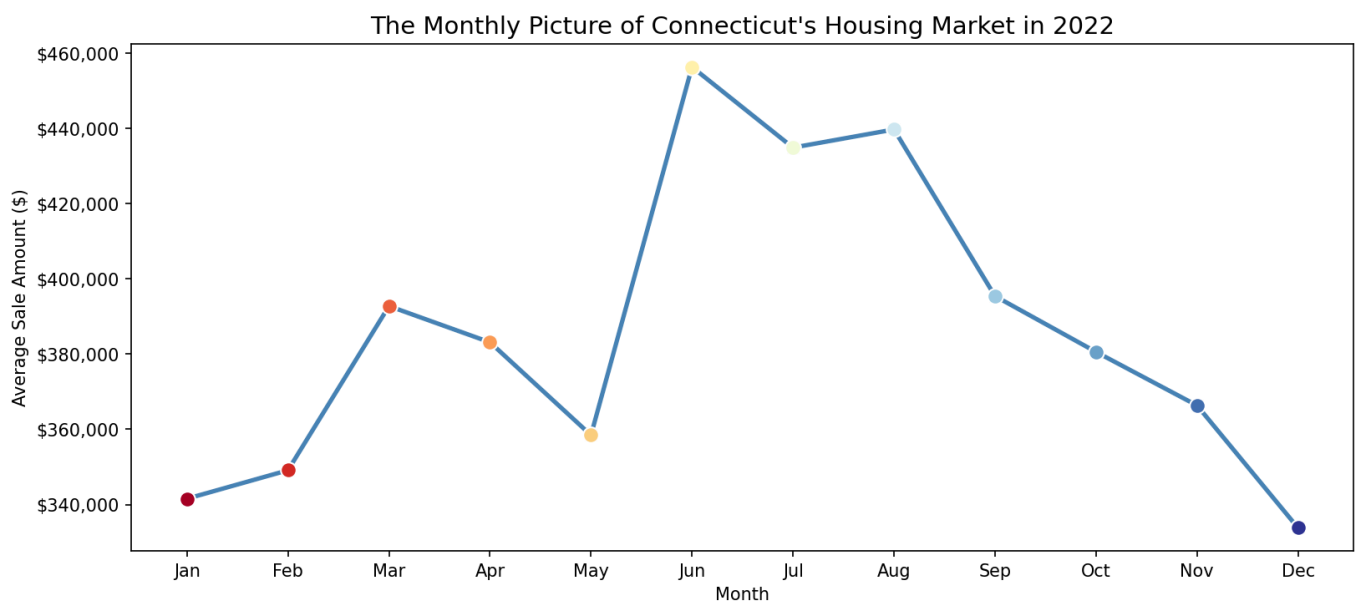
This chart shows the 10 towns in Connecticut with the highest average sale prices in 2022. Westport came out on top at \$1,198,031 average, followed by New Canaan at \$1,165,803 and Weston at \$1,097,308. What really stood out to me was that even the cheapest town on this list, Salisbury, still averaged \$848,167 so every single town in the top 10 was over \$800K. Most of these towns are in Fairfield County which is the part of Connecticut closest to New York City so it kind of makes sense that prices are that high there.

Figure 4 - Towns With the Most Sales



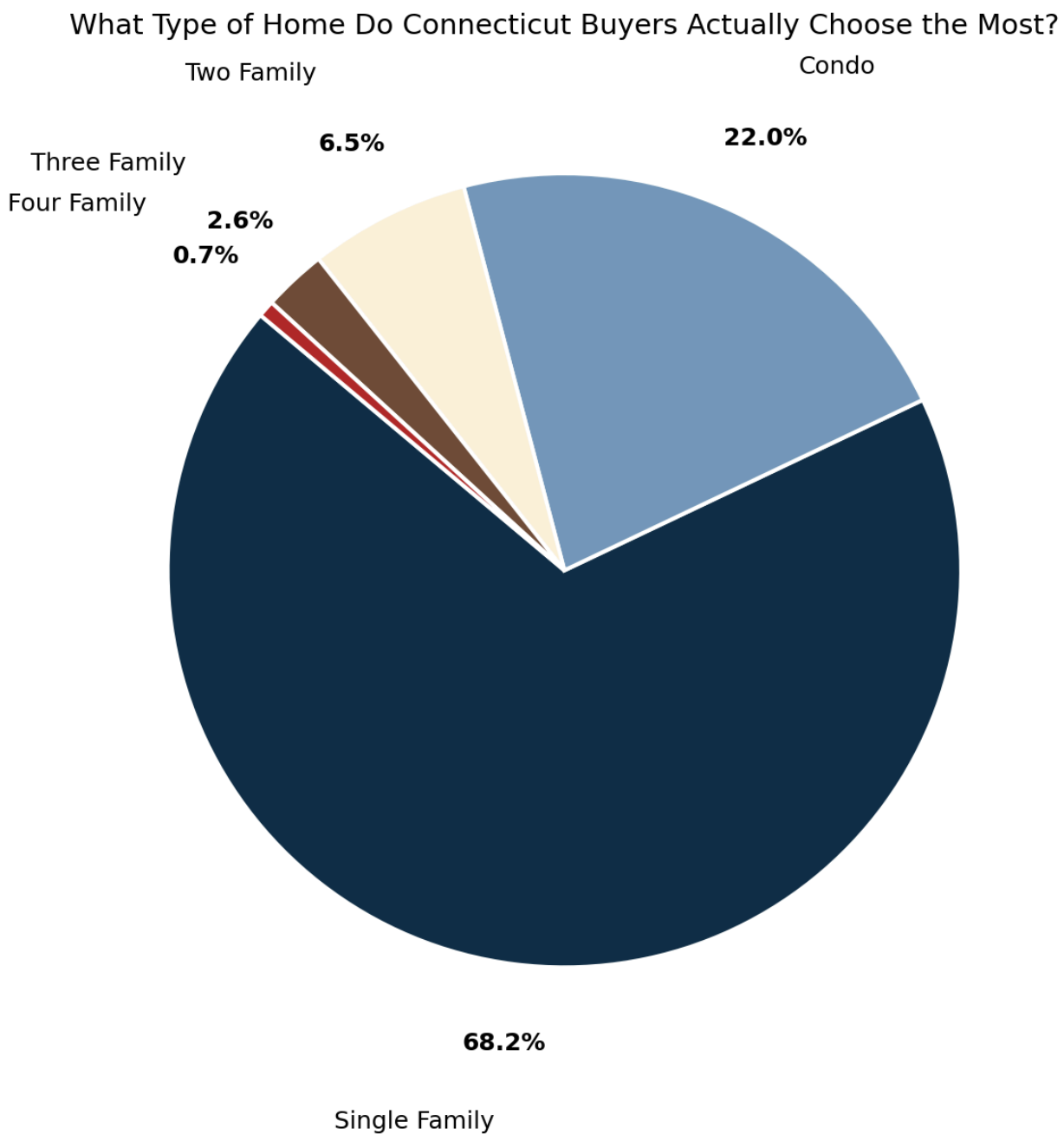
This chart shows the 10 towns where the most homes were sold in 2022. Waterbury had the most with 165 sales, followed by Stamford with 150 and Bridgeport with 136. The thing that really stood out to me was that none of these towns showed up in the top 10 most expensive list at all. So the towns where people are actually buying the most homes are completely different from the towns with the highest prices which makes a lot of sense when you think about it because not everyone can afford to buy in Westport or New Canaan.

Figure 5 - Average Price by Month



This chart shows how the average home sale price changed each month throughout 2022. January and December were the cheapest months at around \$340K. June had this huge spike up to around \$455,000 which was way higher than every other month, which is what I actually expected because of the summer time. After June it kind of drops back down and just keeps going lower through the rest of the year.

Figure 6 - Sales by Residential Type



This pie chart shows what types of homes were sold the most in 2022. Single family homes made up 68.2% of all the sales which is way more than any other type, so almost 7 out of every 10 homes sold in Connecticut was a single family home. Condos were second at 22% and everything else was really small, Two Family at 6.5%, Three Family at 2.6%, and Four Family at

only 0.7%. This tells you that Connecticut is really dominated by single family homes and that is just what most people are buying there.

For the models I trained three of them and compared them to see which one did the best job predicting home prices. To compare them I used two things, R^2 and MSE. R^2 basically tells us how close the model's guesses were to the real prices, kind of like a score out of 1. So if R^2 is 1.0 the model guessed perfectly every time and if it's 0 it basically just guessed randomly. MSE tells us how far off the predictions were on average, so lower is better.

Model	Train R^2	Test R^2	Test MSE
Linear Regression	0.0148	0.0169	82,809,853,786
Polynomial Regression	0.5614	0.5345	39,211,234,193
Decision Tree	0.8480	0.7334	22,459,714,659

Looking at the table, Linear Regression got a Test R^2 of only 0.0169 which is basically a zero so it pretty much failed. Polynomial Regression did way better at 0.5345 so it got about halfway there. The Decision Tree was the clear winner with a Test R^2 of 0.7334 which means it got 73% of the way to a perfect score on brand new data it had never seen, and it also had the lowest MSE meaning its price predictions were the closest to the real sale prices out of all three models.

Figure 7 - Model Comparison

Which Model Predicted Connecticut Home Prices the Best?

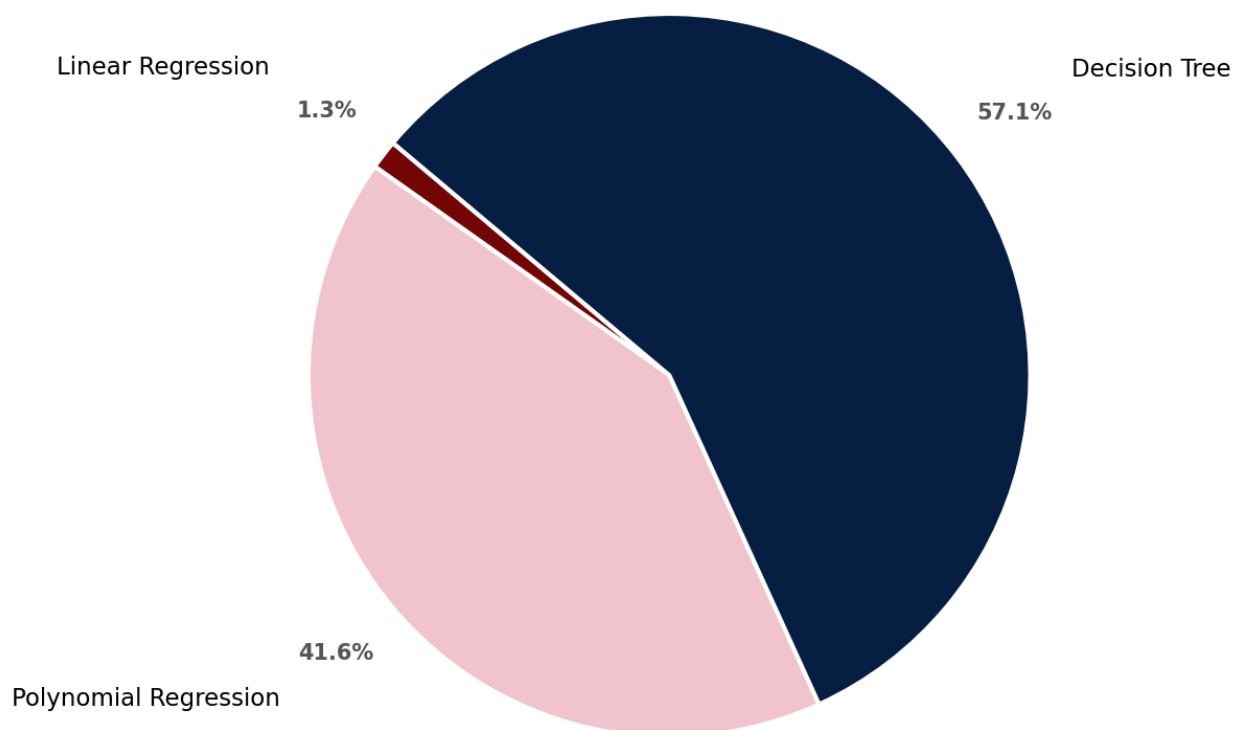


Figure 7 shows the same thing but as a pie chart and you can really see how much bigger the Decision Tree slice is compared to the other two.

I also tested the Decision Tree on a totally made up house just to see if it would give a realistic number. I used a single family home in Hartford with an assessed value of \$150,000 and it predicted the sale price would be \$293,986.05. That actually seems really realistic for Hartford so it was cool to see the model actually work on something concrete.

Discussion:

The main takeaway from this project is that where a home is located in Connecticut matters way more than what type of home it is. Looking at the charts, the gap between towns like Westport at over \$1 million and the towns where people are actually buying like Waterbury is huge. They are basically two completely different housing markets in the same state which I thought was really interesting. The main conclusion I can draw is that location is by far the biggest factor in how much a home sells for in Connecticut. The box plot showed that the type of home did not make that big of a difference on price since all the medians were pretty close to each other, but the town made a massive difference. A home in Westport and a home in Waterbury are not even in the same conversation price wise.

Something that I thought was really interesting and surprising was that the most expensive towns and the towns with the most sales were totally different. You would think popular towns would also be expensive but it is actually the opposite. The towns people are buying in the most are the more affordable ones which makes sense but it was cool to actually see it proven in the data.

The model results also surprised me a little. I honestly did not expect Linear Regression to do that badly. Getting basically a zero means it could not figure out any real pattern in the prices at all. The Decision Tree doing so much better makes sense now because home prices are complicated and depend on a lot of specific things like exactly which town and what type of home, so a model that can make those specific decisions is going to do way better than one that just tries to draw a straight line through everything.

Challenges:

The first challenge I ran into was the Sale Amount and Assessed Value columns. When I loaded the data I assumed they would already be numbers but they came in as strings with dollar signs and commas in them like "\$250,000" so Python could not do any math on them at all. I had to use `str.replace()` twice to strip out both the dollar sign and the comma before converting them with `pd.to_numeric()` which took me a while to figure out.

Another challenge was figuring out what to do with the rows that were missing a Residential Type. My first instinct was to just drop them but that felt wasteful since those rows still had valid sale prices and assessed values. I ended up filling them in with "Unknown" instead so I could keep the data without introducing anything fake into the numbers.

Also making the charts in Spyder was kinda confusing for me at first because I was not sure which types of charts I should make or how they would look and how the text and labels would come out. But I just kept trying different things until they looked the way I wanted them to.

What I Learned:

The main thing I learned from doing this project was how to actually use machine learning models in Python from start to finish. Like in class we talked about how the models work but actually doing it myself on a real dataset was totally different. I had to figure out how to clean the data, encode the categorical columns, split everything into training and testing sets, train three different models, and then actually compare them to see which one did the best. Doing all of that on my own made it click in a way that just learning about it in class did not. I also learned that not every model works for every dataset and that you really have to try different ones to see what actually performs well on your specific data.

Citations

Cates, S. (2026). Course demo codes and lecture materials. COMSC 225.01, Roger Williams University.

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